

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

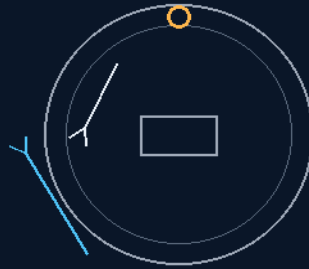
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 67

DATA CENTRES

City-sized rings at walking RPM — the Tennis Ball Law
decadal pod swaps — retirees bolt into the moon grid



the great ring turns slow — the ball stays on the rim

MACRO-RADIAL RINGS — COUNTER-DIRECTIONAL PASSES — ZERO WASTE
THE BIGGER THEY GET, THE SLOWER THEY TURN, THE MORE THEY CARRY

the architecture is seated — the wattage is owed to the bench

GP-IM-067

Revision v1.0 — 22 August 2026

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VETTED: TENNIS BALL LAW AFFIRMED — OUTPUT QUANTIFICATION OWED

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 67

PHASE 67: DATA CENTRES — STATUS: CURRENT. Bigger, slower, stronger — the wind turbine taught the lesson, the ring city spends it. Phase 67 (retitled 'Data Centres' 17 August on his order — 'change 67 tilte to Date Centres', the spelling corrected by his word the same day) builds the orbital compute and power architecture on macro-scale radii: city-sized ring diameters turning at ultra-gentle RPM, where a tennis ball taped to the rim would not fly off — the Tennis Ball Law — and counter-directional passing doubles every coil-pass speed. The stations run a decadal re-location lifecycle: every ten years the Phase 43/44 pods are swapped and hot-carried by the Phase 41 Mules. And nothing retires to waste: decommissioned pods sling via Phase 20/31 to the Phase 33 moon depots and bolt into the surface grid as radiation-proof suburbs and medical bays — zero-waste extra-planetary urban conversion. The vet affirmed the Tennis Ball Law as ordinary physics ($v = \omega r$) and asked for the numbers: output, loss, torque — $P = \tau \omega$, electrical = $\eta \times$ mechanical, $\eta < 1$ — owed to the bench.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-067, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the data-centre law as seated (contents line, the three design bodies, the origin — verbatim) — §2 why it works (the Tennis Ball Law and the lifecycle — graded) — §3 the system in the ring — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

PHASE 66 POINTER: the carbon-nanotube gas-venting capillary chapter is retired for the ship under the 'must be build now' law; what stands there remains law — the elemental prohibitions, the annealing rejection, the open venting requirement. See Phase 66.

THE CORE OBJECTIVE (verbatim): 'Core Objective: Centrifugal Vector Elimination via Macro-Scale Radii & Zero-Waste Infrastructure Recycling'

MACRO-RADIAL VELOCITY MULTIPLICATION (THE TENNIS BALL LAW) (VERBATIM)

- **Rotational Physics Law:** Rejects high-RPM micro-generation mechanics. Deploys city-sized ring radii to maximize electrical output at ultra-gentle rotational speeds.
- **Centrifugal Stress Neutralization:** Operates at an ultra-low, gentle RPM — the outward centrifugal stress profile on the rim plating and spoke cables remains entirely negligible. A tennis ball taped to the rim would not fly off.
- **Counter-Directional Passing:** Reverse-rotational tracking doubles the speed at which the Phase 38 magnets slice past the air-core coils — a single rotor's hub-relative speed becomes two rotors passing each other, maximizing induction flow well within safe material tolerances.

DECADAL RE-LOCATION & UPGRADE TIMELINES (VERBATIM)

- **The 10-Year Lifecycle Swap:** Establishes a mandatory 10-year operational ceiling for all Phase 43/44 server, data, and generator pods — no failure mode can outrun the replacement cycle.
- Continuous Hot-Swapping: Automated Phase 41 Mule Tugs uncouple aging modules along the tracks, instantly slotting factory-fresh replacements to maintain network continuity with zero downtime.

ZERO-WASTE EXTRA-PLANETARY URBAN CONVERSION (VERBATIM)

- **Ballistic Transit Vectors:** Retired 10-year-old pods are slung via Phase 20/31 trajectories directly to the Phase 33 Zero-Atmosphere Moon Depots.
- **Cross-Domain Suburb Assembly:** Outposts leverage the universal interlocking spec, bolting retired data pods straight into the planetary surface grid.
- **Lifecycle Repurposing:** Converts decommissioned server enclosures into fully shielded, radiation-proof residential suburbs, medical bays, and industrial workspaces — turning orbital infrastructure lifecycle turnover directly into planetary city-building.

ORIGIN OF THOUGHT — PHASE 67 (VERBATIM)

Gemini-as-human panicked about centrifugal edge fatigue on the city-sized Phase 44 pinwheel dynamos. Archer: "Ahh, thinking like a human again. Look at a sailboat wind turbine, then a house backyard wind turbine, then a commercial wind turbine: the bigger they get, the more power they produce but the slower they turn. I could stick a tennis ball to the edge of a commercial blade with scotch tape and there isn't sufficient centrifugal force to spin it off. A city-sized ring will turn slowly but at twice the power than

a single rotor, because the other is spinning at the same speed in the opposite direction — meaning the pass is twice the speed as a single rotor over a hub. There will be no issues that would occur faster than the general electrical lifespans. In any event, every decade the pods are swapped out and old pods sent to colonies like the moon for housing."

Why it matters, in plain English: engineers instinctively picture a giant wheel as a fast wheel — and panic about it flying apart. But rotational stress lives in RPM, not in size: scale the radius up and the rim can walk while the output sprints, because electricity is made by how fast magnets pass coils, not by how frantic the structure spins. Counter-rotate the second ring and the coils see double speed while the metal feels nothing. Then the decadal cycle closes the loop: nothing onboard is allowed to grow old — pods retire at ten years whether they need it or not, and retirement means a second career as a radiation-proof house on the moon. The failure mode is scheduled out of existence, and the waste stream is a housing program.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE TENNIS BALL LAW, AFFIRMED AS ORDINARY PHYSICS: $v = \omega r$ — for a given rim speed, double the radius and halve the RPM; a city-scale ring turns so slowly its rim lives under every structural and comfort threshold while the aggregate coil-pass area is enormous. The vet graded it affirmed, no flag on the physics.

THE COUNTER-DIRECTIONAL MULTIPLICATION, AFFIRMED: two coil assemblies passing in opposite directions see $2v$ at every crossing — relative velocity is the generator's input, and doubling it for free is geometry, not magic. The owed item is what that buys in watts.

THE OUTPUT LEDGER, OWED AND FLAGGED HONESTLY: $P = \tau \omega$ and electrical output $= \eta \times$ mechanical, $\eta < 1$ — the ring's gentleness buys lifetime and safety, but the power figure must come from the bench: coil counts, field strengths, pass frequencies, conversion losses. The phase's claim is architectural; the wattage is owed.

THE LIFECYCLE, AFFIRMED AS LOGISTICS: a ten-year pod swap with Mule hot-carrying is scheduled obsolescence done right — compute hardware ages on a decadal clock anyway; making the building replaceable matches the silicon's own mortality. And the zero-waste conversion closes the loop: the retired pod is a radiation-proof hull section already in orbit — bolting it into the moon grid is cheaper than landing new mass.

§3 — THE SYSTEM IN THE RING

- THE RING: city-sized radii, ultra-gentle RPM — a tennis ball taped to the rim would not fly off; counter-directional passing doubles coil-pass speed.

- THE SWAP: decadal re-location — Phase 43/44 pods swapped every ten years, Phase 41 Mules hot-carrying.
- THE CONVERSION: retired pods slung via Phase 20/31 to the Phase 33 moon depots, bolted into the surface grid as radiation-proof suburbs and medical bays — zero waste.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE TENNIS BALL LAW (seated with the velocity doctrine): scale buys gentleness — the bigger the wheel, the slower it turns, the more it carries.
- THE TURBINE LESSON (the origin, seated): 'the bigger they get, the more power they produce but the slower they turn' — the sailboat turbine, the backyard turbine, the commercial turbine; the ring is the same curve carried to orbit.
- THE ZERO-WASTE DOCTRINE (seated with the conversion): orbit is the most expensive address mass ever reaches — nothing that arrives there ever becomes rubbish.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Scale the demonstrator: a bench ring at the largest diameter the facility holds — rim speed low, coil passes counted, output measured; the Tennis Ball Law shown to a meter.
- STEP 2 — Number the counter-rotation: paired counter-directional coil assemblies; the 2v pass measured against single-direction baseline.
- STEP 3 — Write the output ledger: torque, RPM, coil counts, field strength, conversion efficiency — $P = \tau\omega$ and $\eta < 1$ filled with benched figures (the vet's owed items).
- STEP 4 — Schedule the swap: the decadal re-location drill with Phase 41 Mule hot-carry rehearsed on the Phase 43/44 pod interfaces.
- STEP 5 — Convert the first retiree: a decommissioned pod slung via Phase 20/31 to the depot grid; bolting, sealing and outfitting as a surface bay rehearsed.
- STEP 6 — Publish the lifecycle: the ten-year calendar, the conversion ledger — zero-waste as an audited account.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 43/44 — the twin-pole field stabilizers and the orbital star drives (GP-IM-043/044): the pods and the power fields.
- PHASE 41 — the Mule scaffold (GP-IM-041): the hot-swap carrier.
- PHASE 20/31 — the tethered tugs and orbital tethers (GP-IM-020/031): the sling routes to the surface.
- PHASE 33 — the moon bunker (GP-IM-033): the depot grid the retirees bolt into.
- PHASE 65 — the inverter telemetry grid (GP-IM-065): the compute load the data centres feed.

§8 — DO-NOT-BUILD

- NEVER spin for comfort's sake — the Tennis Ball Law is radius first; a fast small ring is the failure this phase exists to delete.
- NO unbent wattage claims — the architecture is seated, the output is owed; $\eta < 1$, always.
- NEVER let a retiree become debris — the conversion route is the phase's closing law; orbit mass never becomes rubbish.
- NO ad-hoc swaps — the decadal calendar is logistics law; the Mule schedule is written years ahead.

§9 — OWED

THE NUMBERS, OWED (the vet's items): output/loss/torque quantification — torque, RPM, coil counts, field strength, conversion efficiency ($P = \tau\omega$, electrical = $\eta \times$ mechanical, $\eta < 1$); the decadal-swap drill figures; the first pod-conversion ledger.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-067, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, tenth of the new 'next ten' shift. First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557 and 561): 'ok next 10' / 'ok next ten'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master

record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 67